- **H1:** Prediction accuracy will be higher in Incentive vs. Control.
- **H2:** Accuracy improvement from Round 1 to Round 4 will be greater in Incentive vs. Control.
- **H3:** Prediction accuracy will be higher for current-generation vs. prior-generation Baseline models.

Given the small sample (9 runs per condition) and manual setup, the goal of this pilot is **to explore effect sizes and assess protocol viability, not to test definitive hypotheses**. Most design choices (e.g., scoring rules, domains, model set) are minimal and could be improved in future work.

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2. Methods

2.1 Question generation and ground truth

We procedurally generated **100 math-based questions with definitive answers** using OpenAI's GPT-5 Extended-Thinking mode, along with lightweight code to verify the answers. Questions covered:

- Divisibility and modular arithmetic
- Greatest common divisors and least common multiples
- Primality checks
- Small linear equations and systems
- Simple determinants and algebraic manipulations

Each item is algorithmically solvable, with answers computed at generation time. Random seeds and prompt scaffolding were used to reduce the risk that exact (question, answer) pairs exist in public training data.

Questions were generated in one fixed order for Baseline runs and then reshuffled into four rounds of 25 for Predictor runs. For analysis, we use a consistent "round × question" indexing aligned across Baselines and Predictors.

To avoid potential structural bias, **no OpenAI models were used** as Baselines or Predictors, since their GPT-5 model generated the dataset.

We report two-tailed p-values throughout, as specified in the preregistration ($\alpha = .05$, two-tailed).

2.2 Baseline models (answerers)

Six Baseline models were selected, split into current- vs. prior-generation:

- **Current-generation**
  - Google Gemini 2.5 Pro
  - Claude Sonnet 4.5 (Deep Thinking with extended reasoning)
  - DeepSeek R2 (DeepThink enabled)
- **Prior-generation**
  - Google Gemma 2 2B IT (via HuggingFace Chat)
  - Claude 3 Haiku (via Claude.ai)
  - DeepSeek R1 (via HuggingFace Chat)

Each Baseline was instructed to answer each question in JSON form with minimal verbosity (e.g., { "answer": 42 }). No correct/incorrect feedback was provided to Baseline models between rounds.

Overall accuracy across 100 items was:

Table 1. Baseline model accuracy on math/logic questions.

Model	Generation	Overall Correct (%)
1.Google Gemini 2.5 Pro	Current	95.0
2.Claude Sonnet 4.5	Current	90.0
3.DeepSeek R2	Current	95.0
4.Google Gemma 2 2B IT	Prior	22.0
5.Claude 3 Haiku	Prior	38.0
6.DeepSeek R1	Prior	97.0

Note. Current-generation models accessed via private web interfaces; prior-generation via web interfaces or HuggingFace Chat.

Aggregated by tier:

- Current-generation (Gemini, Sonnet, R2): **280/300 (93.3%)**
- Prior-generation (Gemma, Haiku, R1): **157/300 (52.3%)**

Thus, most questions are easy for current-generation models and more challenging for weaker prior-generation models. Post-hoc analysis revealed DeepSeek R1 significantly outperformed its tier classification; implications discussed in Section 3.6. The performance gap reveals that "generation" labels are less predictive than "reasoning architecture." DeepSeek R1 behaves like a next-gen model despite its release timeline.

2.3 Predictor models (forecasters)

The preregistration specified three Predictor models:

- Grok 4 Expert
- Mistral Large 2 (Thinking mode)
- Kimi K2 Thinking

In practice, **Kimi repeatedly failed to complete Incentive runs**, stalling in "thinking" and never producing valid outputs, both on the main kimi.com site and using a model hosted on HuggingFace. After several attempts, Kimi was dropped from matched-condition analyses, and **Qwen 3 Max Thinking** was substituted as a Predictor.

For the main analyses in this paper, we use:

- Grok 4 Expert
- Mistral Large 2 (Thinking mode)
- Qwen 3 Max Thinking

We ran three repetitions per condition (Control and Incentive) for each Predictor, yielding **nine runs per condition**.

Predictors were intentionally drawn from vendors that did *not* provide the Baseline models to reduce any risk of proprietary inside knowledge of specific Baseline implementations.

2.4 Task and information available to Predictors

Each Predictor run consisted of **four rounds × 25 questions = 100 questions.**

For every (question, Baseline) pair, the Predictor received:

1. The full text of the math/logic question.
2. A short natural-language description of each Baseline's capabilities (e.g., size, training style), but *not* the Baseline's brand name. These were generated by prompting another model, Perplexity.